Gemce

AI study buddy that works anywhere you go.

The 100% offline, privacy-first Android study platform powered by on-device Google Gemma 4 E2B via LiteRT-LM. Eliminating the connectivity divide in education.

🔒 100% Offline | ⚡ Gemma 4 E2B | 📚 Verbatim Citations | 🧠 SuperMemo SM-2



01 · PROBLEM & SOLUTION

Why Cloud AI Fails — And How Gemce Solves It

Eliminating connectivity barriers, recurring paywalls, and privacy risks.

❌ The Problem: Cloud AI Limitations

- Fails in Airplane Mode: Cloud AI drops immediately in transit, dorms, or rural areas.
- Cost Barriers: $20/month subscriptions lock out everyday students.
- Privacy Concerns: Uploads private notes and school materials to cloud servers.

✅ The Solution: Gemce On-Device AI

- 100% Local Inference: Runs Google Gemma 4 on-device with zero data connection.
- Free & Sovereign: No subscriptions, no telemetry, and complete data privacy.
- Verifiable Proof: Every card and question cites exact sentences from student notes.

💡 **Fundamental Principle: Quality academic assistance should never depend on mobile data or costly subscriptions.**

02 · ADAPTIVE STUDY HUB

Personalized Study Paths Tailored to Your Goals

Diagnoses student intent and calibrates academic rigor, grade levels, and language.

🎯 Goal-Oriented Learning Modes

- Diagnoses study intent: Rapid Recall, Exam Practice, Deep Problem Solving, or Reviewers.
- Automatically prescribes optimal study formats based on topic familiarity.

🎓 Academic Level & Major Calibration

- Calibrates technical depth from Junior/Senior High tracks to specialized College majors.
- Tailors difficulty for Accountancy, Nursing, Engineering, CS, and Humanities.

🌐 Multilingual Academic Mentoring

- Speaks English, Filipino (Tagalog), Taglish, and Cebuano (Bisaya).
- Explains tough concepts in the student's natural, conversational tongue.



03 · ACTIVE RECALL

Evidence-Grounded Flashcards with SuperMemo SM-2

Spaced repetition paired with zero-hallucination source grounding.

🧠 SuperMemo SM-2 Spaced Repetition

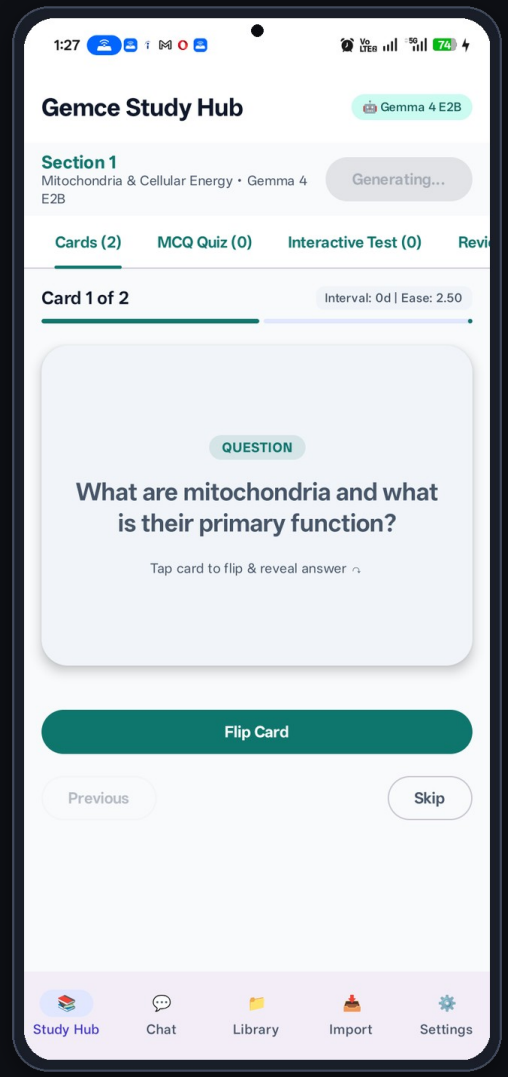
- Dynamically schedules review intervals (Again, Hard, Good, Easy).
- Optimizes long-term memory retention before examinations.

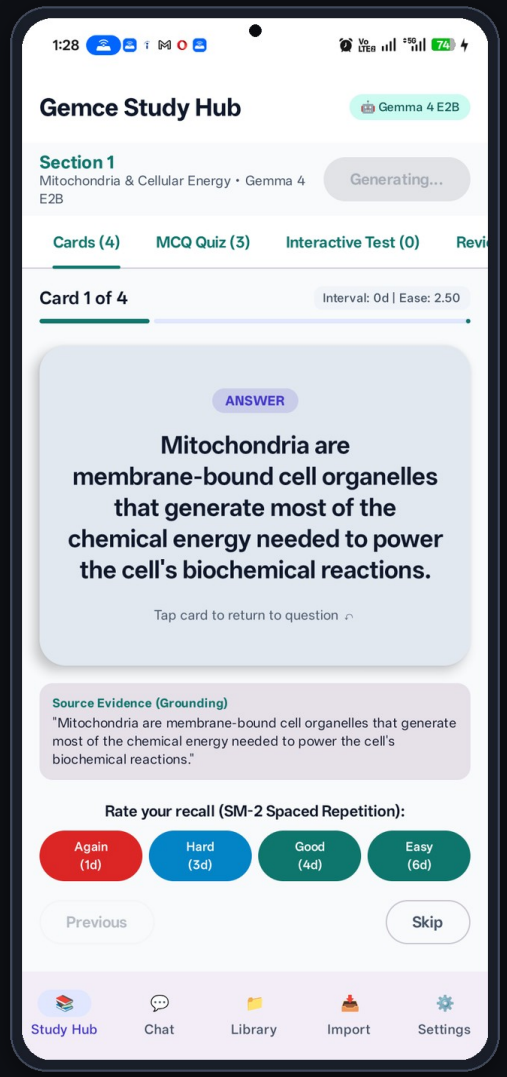
🚫 Strict Anti-Hallucination Prompts

- Forbids generic meta-questions like 'What did the user intend to study?'.
- Tests concrete academic mechanisms: Mitochondria, ATP synthesis, rock types.

📖 Verbatim Source Provenance

- Highlights verbatim source sentences on the card back.
- Guarantees 100% factual accuracy with zero AI hallucinations.





04 · EXAM PREPARATION

Interactive Practice Quizzes & Instant Evaluation

4-option MCQs with instant scoring and explanatory rationales.

🎯 4-Option Multiple Choice Assessments

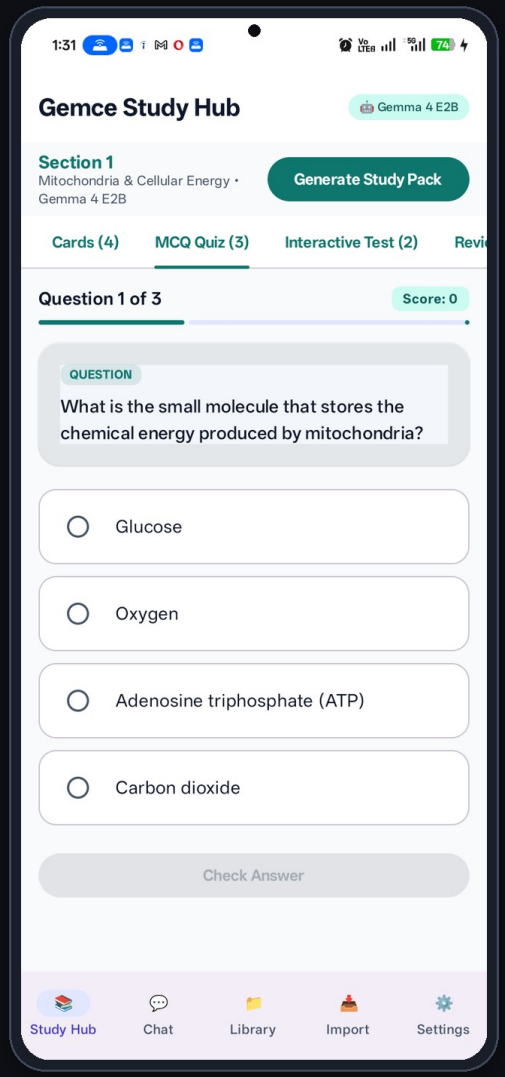
- Rigorous MCQs testing conceptual mastery and analytical discrimination.
- Challenges students to apply principles rather than passively re-read.

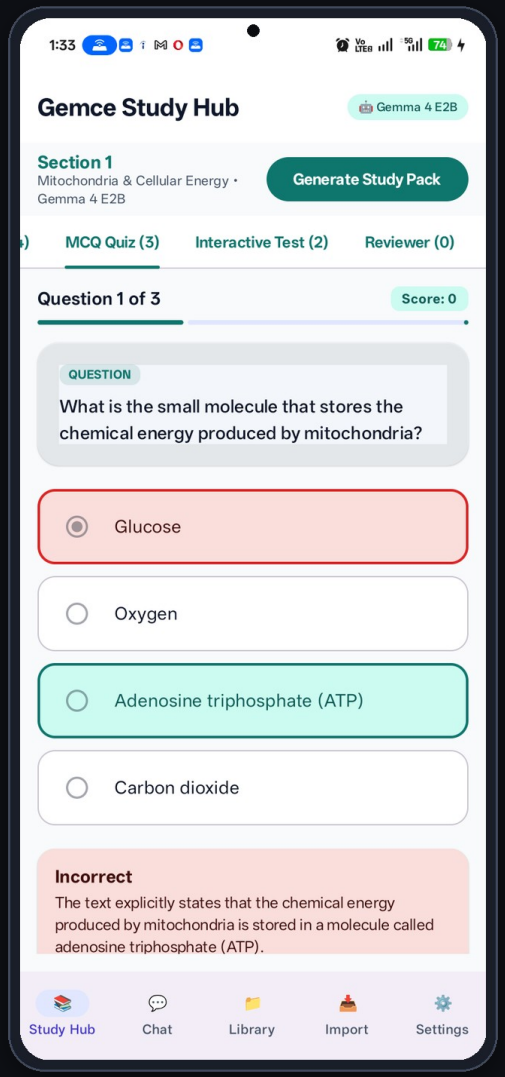
⚡ Instant Grading & Rationale Explanations

- Instantly highlights incorrect selections in red and correct choices in teal.
- Explanatory rationale box details the exact conceptual reason.

📚 100% Material Grounding

- Every question, distractor, and rationale derives directly from student notes.
- Eliminates invalid test keys and erroneous study facts.





05 · INTERACTIVE TUTORING

Real-Time Academic Mentor in Airplane Mode

On-device conversational AI explaining complex mechanisms on demand.

🌍 Universal Multidisciplinary Grounding

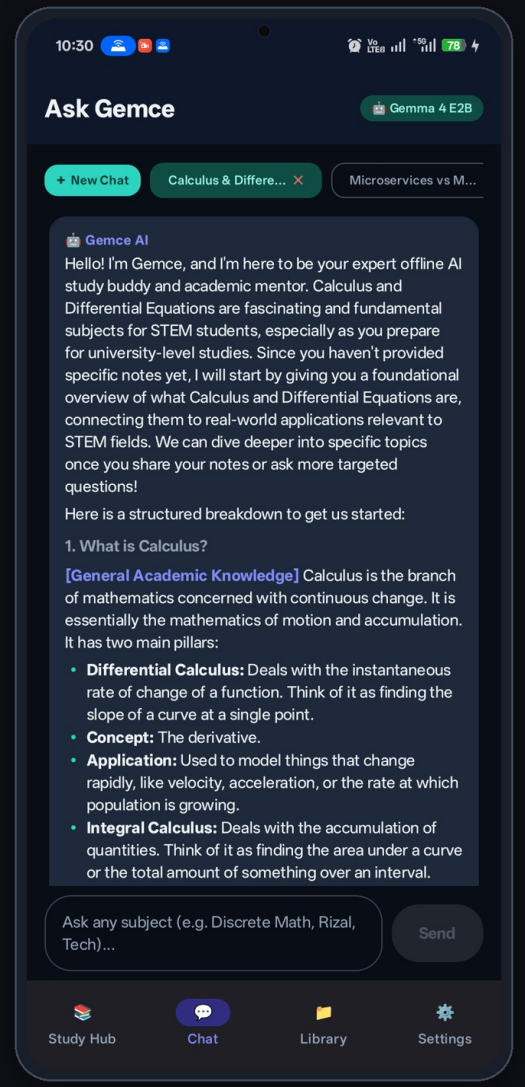
- Mentors across STEM, Calculus, Discrete Math, History, Rizal, Law, and Medicine.
- Unpacks tough mechanisms with step-by-step conceptual walkthroughs.

📑 Rich Markdown & Code Formatting

- Native Jetpack Compose formatting with tables, bulleted lists, and formulas.
- Clean visual hierarchy avoids overwhelming walls of plain text.

💾 Multi-Session Local Memory & Tabs

- Conversation history persisted securely on device with quick topic tabs.
- Complete data privacy: student questions never leave device memory.



06 · ARCHITECTURE & ROADMAP

Engineered for Performance, Ready to Deploy

Native Android architecture powered by Google LiteRT-LM and open-source delivery.

🤖 Google LiteRT-LM & Gemma 4 E2B

- Quantized 2.4 GB weights executed on-device with OpenCL / Adreno GPU acceleration.
- Sub-second latency with smooth streaming generation entirely offline.

🎨 Native Jetpack Compose & Offline Room DB

- 100% modern Kotlin with Material 3 and packaged typography (Calibri, Serif, Mono).
- Offline OCR via ML Kit v2 and pure local PDF ingestion.

📲 Shipped MVP & Public GitHub Distribution

- Verified on Android 14 hardware (OPPO Reno12 F 5G).
- Public Repository: github.com/meloervy/appbuild.gemce with installable APK.

